

ARCHITECTURE DESIGN DOCUMENT

Retail Analytics Platform on Microsoft Azure

Option A — Big Data Architecture Project

Prepared for: Enterprise Architecture Review Board

Classification: Internal — Confidential

Date: April 2026

Budget Ceiling: \$25,000 / month

Executive Summary

This document presents a production-ready **Azure big-data architecture** for a mid-size retail chain operating **150 stores**. The solution ingests four distinct data streams — point-of-sale transactions, loyalty-program records, real-time inventory feeds, and social media mentions — and transforms them into daily sales reports, real-time stockout alerts, weekly customer-segmentation models, and monthly demand forecasts. The estimated monthly infrastructure cost of **~\$18,795** sits comfortably within the \$25,000 ceiling, leaving a 25 % buffer for growth and experimentation.

Section 1 — Data Sources & Ingestion Layer

The platform ingests data from four sources, each requiring a different ingestion strategy based on its volume, velocity, and latency tolerance.

Data Source	Volume / Freq.	Ingestion Method	Azure Service
POS Transactions	500K / day	Batch (nightly)	Azure Data Factory
Loyalty Database	100K members	Batch (daily sync)	ADF + Azure SQL DB
Inventory Updates	Real-time, 150 stores	Streaming	Azure Event Hubs
Social Media Text	~1,000 / day	Batch (hourly)	ADF + Logic Apps

Point-of-sale transactions (500,000 records/day, ~350 MB compressed CSV/JSON) are non-latency-sensitive at the ingestion stage because the business requirement is a 7 AM report, not a live feed. **Azure Data Factory (ADF)** pipelines triggered at 11 PM pull data from store SQL endpoints, apply schema validation, and land files in the raw zone of ADLS Gen2. Batch ingestion is the correct choice here: it minimises per-transaction overhead, enables bulk compression, and costs a fraction of streaming equivalents.

Inventory updates from 150 stores require near-real-time processing because stockout alerts must fire within minutes. Each store publishes JSON messages over HTTPS to **Azure Event Hubs** (10 Throughput Units, Standard tier), which buffers up to 1 MB/s per TU — well above the expected ~500 KB/s peak load. Event Hubs decouples producers from consumers and guarantees at-least-once delivery.

Loyalty data is a structured relational database with low daily churn (~5 %). A nightly ADF copy activity performs an incremental sync using watermark columns (last_modified_at), avoiding full-table scans. **Social media text** is collected hourly via Logic Apps calling the platform APIs, then handed to ADF for staging — batch is sufficient given the low volume and the weekly cadence of sentiment analysis.

Section 2 — Storage Layer

All persistent data resides in **Azure Data Lake Storage Gen2 (ADLS Gen2)**, organised into three medallion zones. ADLS Gen2 is chosen over Azure Blob Storage because it provides POSIX-compatible ACLs, hierarchical namespace for efficient directory operations, and native integration with Synapse Analytics and Azure Databricks — eliminating costly data copies.

ADLS Gen2 Zone	Contents	Format	Retention
Raw (Bronze)	All source data as-is	CSV / JSON / Parquet	90 days
Processed (Silver)	Cleansed, deduplicated	Delta Parquet	1 year
Curated (Gold)	Aggregates, KPIs, ML features	Delta Parquet	3 years

Partitioning strategy: Transaction data is partitioned by **store_id / year / month / day** to support predicate push-down in Synapse SQL pools. Inventory snapshots are partitioned by **store_id / date**. This organisation reduces scan size by >90 % for single-store or single-day queries, directly lowering compute costs.

Real-time inventory state — the latest stock level per SKU per store — is also materialised in **Azure Cosmos DB (Core SQL API)**. Cosmos DB is selected for this hot-path read store because it offers single-digit millisecond reads at any scale, supports automatic indexing on SKU and store dimensions, and integrates natively with Stream Analytics outputs. The transactional ADLS landing zone covers durability and history; Cosmos DB covers operational speed.

Lifecycle management policies automatically tier ADLS blobs from hot to cool storage after 30 days and to archive after one year, reducing storage costs by an estimated 40 % compared with keeping all data on hot tier.

Section 3 — Processing Layer

Processing is split between a **real-time path** for inventory alerts and a **batch path** for all reporting and ML workloads.

Real-time path — Azure Stream Analytics: A Stream Analytics job (8 Streaming Units) consumes the Event Hubs inventory stream and applies a tumbling window query: if any SKU's stock_level falls below its reorder_point in a 5-minute window, it writes an alert to Cosmos DB and publishes to an Azure Service Bus topic. Downstream applications (store managers' mobile app, warehouse system) subscribe to this topic. Stream Analytics is preferred over Spark Structured Streaming for this workload because it requires no cluster management, scales SUs independently, and has built-in Event Hubs connectors with exactly-once semantics.

Batch path — Azure Synapse Analytics: ADF orchestrates a dependency-aware pipeline DAG each night. Step 1 (00:30): ADF copy activities land raw files. Step 2 (01:00): Synapse Spark pools run data-quality checks and Silver-zone transformations (deduplication, type casting, join to loyalty dimension). Step 3 (03:00): Synapse SQL pools compute Gold-zone aggregates (daily sales by store/category, customer RFM scores). Step 4 (05:30): Power BI dataset refresh triggered via REST. Reports are ready by 06:50, meeting the 7 AM SLA with a 10-minute buffer.

Machine Learning — Azure Machine Learning: Monthly demand-forecasting models (Prophet + LightGBM ensemble) are retrained on the first day of each month using AML pipelines triggered by an ADF schedule. Inference runs weekly. AML managed endpoints expose a REST API consumed by the replenishment microservice.

Section 4 — Analytics & Output Layer

Data warehouse schema: The Gold zone follows a **star schema** in the Synapse dedicated SQL pool: a central **fact_sales** table (transaction_id, store_id, product_id, customer_id, date_id, quantity, revenue) surrounded by **dim_store**, **dim_product**, **dim_customer**, and **dim_date**. This schema is optimised for columnar storage (COLUMNSTORE INDEX) and distribution by store_id, enabling sub-second aggregation queries across all 150 stores.

Visualization — Power BI Premium P1: Four dedicated Power BI workspaces serve different audiences: (1) Executives — daily P&L dashboard, (2) Store Managers — real-time inventory heat-map fed from Cosmos DB via DirectQuery, (3) Marketing — weekly customer-segmentation explorer with RFM cluster drill-through, (4) Supply Chain — monthly demand-forecast vs. actuals. Premium P1 capacity is chosen to enable paginated reports, larger dataset sizes (up to 400 GB), and unlimited viewer licenses — eliminating per-seat costs for 800+ store associates.

API layer — Azure API Management (APIM): Three REST endpoints are published through APIM: /inventory/status (reads Cosmos DB, <10 ms P99), /forecast/{sku} (calls AML endpoint), and /customer/{id}/segment (reads Gold Parquet). APIM enforces OAuth 2.0 via Azure Active Directory, rate-limits per consumer, and logs all calls to Azure Monitor.

Security & compliance: All data at rest is encrypted with Microsoft-managed keys (AES-256). ADLS Gen2 ACLs enforce role-based access: Data Engineers access Silver; Analysts access Gold read-only. Customer PII fields (name, email, loyalty_id) are masked using Synapse Dynamic Data Masking for non-privileged roles. Azure Purview provides data lineage and catalog for GDPR audit readiness. Network traffic is restricted to a private VNet with no public endpoints.

Section 5 — Cost Analysis & Justification

All estimates below are derived from the **Azure Pricing Calculator** (East US region, pay-as-you-go, April 2026 rates) and reflect production-level configurations:

Azure Service	Monthly Estimate	Notes
Azure Data Factory	~\$1,800	40 pipelines, 200K activity runs
Azure Event Hubs (Standard)	~\$700	10 TUs, 30-day retention
ADLS Gen2 (50 TB)	~\$1,000	LRS redundancy, all zones
Azure Synapse Analytics	~\$4,500	DW500c pool, ~600 DWU-hrs/mo
Azure Stream Analytics	~\$800	8 streaming units
Cosmos DB (inventory)	~\$1,200	5,000 RU/s provisioned
Azure Machine Learning	~\$2,500	Forecast model training + inference
Power BI Premium P1	~\$4,995	Embedded capacity, unlimited viewers
Azure API Management	~\$700	Developer tier, 1M calls/mo

Azure Service	Monthly Estimate	Notes
Networking / Monitor / KeyVault	~\$600	Egress, diagnostics, secrets
TOTAL ESTIMATED COST	~\$18,795 / mo	~75% of \$25K budget

The total of **~\$18,795/month** is **25 % below the \$25,000 ceiling**, providing headroom for traffic spikes, new store on-boarding, or A/B experiments. Three optimisation levers hold the budget down: (1) **Reserved Instances** — committing to 1-year reservations for Synapse and AML compute would reduce those line items by ~35 %, saving an additional ~\$2,500/month; (2) **ADLS lifecycle tiering** automatically moves cold data to archive at \$0.002/GB vs. \$0.018/GB hot, cutting storage costs ~40 % at scale; (3) **Synapse serverless SQL** is used for ad-hoc analyst queries (billed per TB scanned) instead of always-on dedicated pools, avoiding idle-compute waste during off-peak hours.

Power BI Premium P1 at \$4,995/month appears steep but replaces ~800 per-user Pro licenses (\$10/user = \$8,000/month), delivering a **net saving of ~\$3,000/month** while unlocking paginated reports and larger model sizes required for monthly demand forecasting outputs.

Architecture Summary & Key Trade-offs

Dimension	Decision	Trade-off Accepted
Latency (inventory)	Event Hubs + Stream Analytics	Higher cost than batch-only
Storage cost	ADLS Gen2 medallion + lifecycle tiers	Extra ETL complexity
Query performance	Dedicated SQL pool (DW500c)	Reserved capacity cost
Operational overhead	Managed PaaS services throughout	Vendor lock-in to Azure
Viewer licensing	Power BI Premium P1 flat fee	Larger upfront commitment

This architecture prioritises operational simplicity and managed services to minimise the internal DevOps burden, while meeting all stated latency, reporting, and budget requirements for the retail analytics platform.